Relational Actualism: Time, Causality, and the Structure of Physical Reality

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Abstract

Relational Actualism is a theoretical framework for the nature and evolution of the physical universe—from fundamental particles to biological complexity—grounded entirely in the reality of time and causality. Its single ontological commitment is that irreversible events are the primitive physical reality: not events *in* spacetime, but the events from which spacetime, matter, and the laws governing them all emerge. From this commitment, and with no free parameters, the framework produces a tightly constrained web of results spanning quantum mechanics, general relativity, particle physics, and cosmology. These include the fine structure constant to seven significant figures, the Standard Model gauge groups, the structural vanishing of the cosmological constant, and the dissolution of the measurement problem. Crucially, these results are not independent findings arranged in a deductive hierarchy. They form a *constraint web*—five interlocking loops of mutual implication in which every major result constrains every other. The theory has no independently tunable parts; a single experimental failure anywhere breaks the entire web. The Bianchi identity, Lorentz invariance, and gauge symmetries are shown to be emergent descriptions—what the discrete, irreversible process looks like when approximated by continuous, time-symmetric mathematics. This reframing identifies every major unsolved problem in fundamental physics as an artifact of that approximation. Ten falsifiable predictions are stated, several testable within five years.

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1 Introduction: A Different Kind of Theory

Physics has long organized its theories as deduction trees: axioms at the top, theorems branching downward, predictions at the leaves. Change an axiom and one branch changes; the rest of the tree is unaffected. Each branch can be evaluated independently.

Relational Actualism is not a tree. It is a *web*: every major result constrains, and is constrained by, every other. The coupling constants constrain the actualization threshold, which constrains the measurement criterion, which constrains the coupling constants. The conservation law constrains the dynamics, which produce the graph on which the conservation law holds. The discrete symmetry constrains the geometry, which constrains the dynamics, which has the discrete symmetry.

This structure is not an accident of presentation. It is the theory. A tree-structured theory can lose a branch without affecting the rest. A web-structured theory cannot—every node

supports every other. This is what makes RA distinctively falsifiable: a single experimental failure anywhere breaks the entire web.

The purpose of this paper is to present RA as it actually is: not as a collection of independent results that happen to share an ontological commitment, but as a single self-consistent constraint web from which all results follow simultaneously.

1.1 The meta-insight

The deepest claim of RA is not any individual result but a structural observation: *every major unsolved problem in fundamental physics is an artifact of approximating a discrete, irreversible process with continuous, time-symmetric mathematics.*

The measurement problem arises because the Schrödinger equation is continuous and time-symmetric while collapse is discrete and irreversible. The cosmological constant problem arises because quantum field theory sums over infinitely many continuous vacuum modes. The incompatibility of general relativity and quantum field theory arises because both are continuum approximations applied simultaneously in a regime where the continuum approximation fails. The arrow of time is "mysterious" only because differential equations are time-symmetric by construction.

In each case, the "problem" dissolves when the continuum description is recognized as an approximation of an underlying discrete, irreversible process. RA provides that process: the growing causal DAG governed by the BDG dynamics.

1.2 Plan of this paper

Section 2 introduces the three primitive structures (the DAG, the LLC, and the BDG action). Section 3 presents the key results as nodes in the constraint web. Section 4 identifies the five constraint loops that give the web its rigidity. Section 5 shows how continuum physics—the Bianchi identity, Lorentz invariance, gauge symmetries—emerges as an approximation of the discrete structure. Section 6 states the falsifiable predictions. Section 7 reflects on what it means for a physical theory to be a web rather than a tree.

2 The Three Primitives

2.1 The growing causal DAG

The fundamental ontological object in RA is a Directed Acyclic Graph (DAG) that grows by the addition of vertices and directed edges. Vertices represent actualization events—irreversible physical interactions. Directed edges encode causal relations: if vertex u causally precedes vertex v, a directed edge points from u to v. Acyclicity encodes irreversibility: no event can cause itself.

Time is not a background parameter. Time *is* the growing of the graph. The past is the part of the graph that has been written; the future is what has not yet been written; the present is the process of writing the next vertex. The arrow of time is not explained by low-entropy initial conditions—it is encoded in the primitive commitment to irreversibility, since a DAG, by definition, cannot grow backwards.

2.2 The Local Ledger Condition (L01)

[LV]

The most fundamental conserved quantity in RA is the Local Ledger Condition: at every vertex v of the DAG, the total conserved charge flowing in equals the total conserved charge flowing out:

$$\sum_{\text{in}} Q(e) \;=\; \sum_{\text{out}} Q(e) \qquad \text{at every vertex } v. \tag{1}$$

This is conservation as bookkeeping—a local balance condition at every node. It has been formally verified in Lean 4 with zero errors, zero warnings, and zero `sorry` tags.[1] The Graph Cut Theorem (L02)[LV] establishes that the LLC holds independently on each side of any causal severance (topological disconnection of the DAG).

2.3 The BDG action in $d = 4$ (L11)

[LV]

When a new vertex v is added to the causal set C, the Benincasa–Dowker–Glaser (BDG) action increment is

$$S(v, C) \;=\; 1 - N_1 + 9\,N_2 - 16\,N_3 + 8\,N_4\,, \tag{2}$$

where N_k counts the number of previously actualized vertices within k causal steps of v in C. The integers $(1, -1, 9, -16, 8)$ are not fitted parameters: they are determined by the combinatorics of causal intervals in a four-dimensional causal diamond [1].

A vertex actualizes if and only if $S > 0$. The BDG closure theorem (L11)[LV], verified in Lean 4 across 124 extension cases, establishes that in $d = 4$ these five integers produce exactly five topologically stable vertex types, corresponding to the five particle classes of the Standard Model.

2.4 The actualization criterion

At the self-dual density $\mu = 1$ (Planck density), the acceptance probability is

$$P_{\text{acc}} \;=\; P(S > 0 \,|\, \mu = 1) \;=\; 0.548436\,, \tag{3}$$

computed by exact enumeration over the Poisson-CSG.[CV] The actualization threshold is

$$\Delta S^* \;=\; -\log P_{\text{acc}} \;=\; 0.60069 \text{ nats}, \tag{4}$$

the minimum quantum relative entropy increase required for an interaction to become a permanent vertex in the DAG. This is not a free parameter—it is computed from the BDG integers and the dimension $d = 4$.

[1] `local_ledger_condition` in `RA_D1_Proofs.lean`.

3 Key Results as Nodes in the Web

3.1 The fine structure constant (IC30)

[CV]

The electromagnetic coupling constant is derived from two independent contributions that enter through a self-consistent Dyson equation:

$$\alpha_{\mathrm{EM}}^{-1} \;=\; 137 + P_{\mathrm{acc}} \cdot c_2 \cdot \alpha_{\mathrm{EM}}\,. \tag{5}$$

The bare coupling $\alpha_0^{-1} = 137$ arises from BDG depth combinatorics: the vertex-type ratio $P(\text{electron})/P(\text{photon}) = \lambda_3\lambda_4 = 1/(3! \cdot 4!) = 1/144$, corrected by the BDG confinement depths $L_g + L_q = 3 + 4 = 7$ (L10)[LV], giving $\alpha_0^{-1} = 144 - 7 = 137$.[LV]

The angular screening term $P_{\mathrm{acc}} \cdot c_2$ has a precise discrete origin: $P_{\mathrm{acc}} = 0.548$ is the BDG acceptance probability at $\mu = 1$, and $c_2 = 9$ is the BDG second-layer coefficient—also the number of integer lattice points in the unit 4-ball. The product $P_{\mathrm{acc}} \times c_2 = 4.936 \approx \pi^2/2 = 4.935$ to 0.023%; the near-identity is a *consequence* of discrete BDG dynamics approximating $d = 4$ geometry, not an input.

Solving the quadratic:

$$\boxed{\alpha_{\mathrm{EM}}^{-1} \;=\; \frac{137 + \sqrt{137^2 + 4\,P_{\mathrm{acc}}\,c_2}}{2} \;=\; 137.036019} \tag{6}$$

Experimental value (PDG): 137.035999. Error: 0.00001% (seven significant figures).

Every input—137 (LV), P_{acc} (CV), c_2 (LV)—is computable from the BDG integers $(1, -1, 9, -16, 8)$. No transcendental numbers appear as inputs. No continuous geometry is invoked.

3.2 The strong coupling constant (C01)

[DR]

The strong coupling at the Z-mass scale follows from the BDG identity $\sum c_k = 0$:

$$\alpha_s(m_Z) = 1/\sqrt{c_2\,c_4} = 1/\sqrt{72} \approx 0.11785\,. \tag{7}$$

The PDG value is 0.1180 ± 0.0009; the agreement is 0.13%.

3.3 Standard Model gauge groups from BDG signs (GS02)

[CV]

The BDG coefficients alternate in sign: $(+1, -1, +9, -16, +8)$. This alternation is forced by the inclusion-exclusion principle applied to causal interval counting in $d = 4$. The sign of c_k determines the angular structure at depth k:

- **Positive** c_k (depths 2, 4): depth-k ancestors *increase* the BDG score S. Vertices are encouraged to have many such ancestors, distributed isotropically across all spatial directions. Result: full SU(3) symmetry at depth 4 (colour), scalar U(1) at depth 2 (Higgs remnant).

- **Negative** c_k (depths 1, 3): depth-k ancestors *decrease* S. Vertices are penalized for many ancestors, typically having one dominant connection that picks out a single spatial direction. Three dimensions split $2+1$: two transverse ($\mathrm{SU}(2)_L$) and one longitudinal ($\mathrm{U}(1)_Y$) at depth 1; heavy confinement penalty ($|c_3| = 16$) at depth 3 (gluon confinement).

This sign mechanism is confirmed by exact enumeration: $E[N_k \mid S > 0]$ is strictly greater than $E[N_k]$ for positive c_k and strictly less for negative c_k, for all $k = 1, \ldots, 4$.

The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is thereby determined by counting—specifically, by the inclusion-exclusion structure of causal intervals in $d = 4$.

3.4 Wave function collapse as actualization

A quantum state is any state with $\Delta S(\rho \| \sigma_0)$ below the threshold ΔS^*. Collapse occurs when, through environmental interaction, the relative entropy crosses the threshold and a new permanent vertex is written into the DAG. There is no Heisenberg cut, no separate collapse postulate, and no many-worlds branching. The Schrödinger equation is the continuum approximation valid when actualization density is low; classical mechanics is the approximation valid when it is high. Both are approximations to the same discrete process.

3.5 General relativity from the LLC

The Einstein field equations $G_{\mu\nu} = 8\pi G\, P_{\mathrm{act}}[T_{\mu\nu}]$ with $\Lambda = 0$ are derived from three machine-verified inputs:

1. The LLC (L01)[LV] provides exact conservation at every vertex.
2. Amplitude locality (O01)[LV] ensures the dynamics depends only on the causal past.
3. The BDG closure (L11)[LV] fixes $d = 4$.

The Benincasa–Dowker uniqueness theorem [1] establishes that these three constraints determine the BDG coefficients $(1, -1, 9, -16, 8)$ as the unique local causal set action. In the continuum limit, this action reduces to the Einstein–Hilbert action $\int R\sqrt{g}\, d^4x$, whose variation yields $G_{\mu\nu}$. The cosmological constant vanishes structurally: the actualization projector P_{act} removes unactualized (off-shell) vacuum contributions from the gravitational source.

This derivation replaces the smooth Lovelock theorem with the discrete BDG uniqueness theorem—no smooth manifold is assumed.

3.6 Cosmological predictions

The Hubble tension is predicted to correlate linearly with baryon density along the line of sight; the KBC void prediction is $H_0 \approx 75.9\,\mathrm{km/s/Mpc}$. WIMPs, axions, and sterile neutrinos are categorically prohibited as dark matter candidates (any particle with zero BDG assembly depth contributes zero weight to the metric source). Boltzmann Brains are structurally impossible: fluctuation-assembled complexity requires P_{acc}^N with $N \sim 10^{26}$, giving probability effectively zero.

4 The Five Constraint Loops

The results of Section 3 do not form a deduction tree. They form a constraint web organized around five interlocking loops.

4.1 Loop 1: Conservation ↔ Dynamics

The LLC (L01) constrains the dynamics to be conservative. The BDG dynamics (determined by O01 and L11) selects which vertices actualize. The actualized vertices define the graph on which the LLC holds. The graph structure constrains what the LLC means at each vertex.

4.2 Loop 2: Dimension ↔ Spectrum ↔ Couplings ↔ Gauge Groups

The dimensionality $d = 4$ (L11) determines the BDG integers. The integers determine the particle spectrum (five topology types) and the coupling constants (α_{EM}, α_s). The BDG closure theorem requires exactly $d = 4$ for five types. The couplings are self-consistent at $\mu = 1$ only in $d = 4$. The alternating signs of the BDG integers determine the gauge group $SU(3) \times SU(2) \times U(1)$ (Section 3.3).

4.3 Loop 3: Discrete Symmetry ↔ Continuous Symmetry

Causal invariance (O02)[LV] ensures the quantum measure is independent of the linear extension. In the continuum approximation, this becomes Lorentz invariance (Section 5.2). Lorentz invariance requires the field equations to be covariant, which in $d = 4$ uniquely selects the Einstein–Hilbert action. The BDG action—the unique discrete action in $d = 4$—is causally invariant, completing the loop.

4.4 Loop 4: Threshold ↔ Coupling

The acceptance probability P_{acc} determines the actualization threshold ΔS^*. The threshold determines the measurement criterion. The product $P_{\mathrm{acc}} \cdot c_2$ enters the screening correction to α_{EM}. The coupling α_{EM} determines the actualization rate for electromagnetic interactions. The actualization rate determines which vertices pass the $S > 0$ filter. The filter determines P_{acc}.

4.5 Loop 5: Bianchi ↔ LLC

The LLC at every vertex becomes $\nabla_\mu T^{\mu\nu} = 0$ in the continuum limit. The field equations give $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$. Therefore $\nabla_\mu G^{\mu\nu} = 0$—the contracted Bianchi identity. The Bianchi identity constrains the metric, which defines the causal structure, which is the graph on which the LLC holds.

The rigidity of these five loops is the theory's principal structural feature. A single experimental failure at any node breaks the entire web.

5 Continuous Physics as Emergent Approximation

The central philosophical claim of RA is that continuous physics is an approximation of the discrete DAG, not the other way around. Three examples make this concrete.

5.1 The Bianchi identity is the LLC (O10n)

[DR]

The contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ is standardly presented as a geometric identity holding for any metric on a smooth manifold. In RA, no smooth manifold exists at the fundamental level. The Bianchi identity is not a geometric fact that the DAG satisfies—it is a *description* of what the LLC looks like when the DAG is approximated as a smooth $d = 4$ manifold.

The analogy is exact: $PV = nRT$ is not a law that individual molecules satisfy. Molecules conserve momentum at every collision. The ideal gas law is what that conservation looks like when you average over 10^{23} collisions. Similarly, the Bianchi identity is what the LLC looks like when you average over 10^{122} actualization events.

5.2 Lorentz invariance is causal invariance (O11)

[DR]

Lorentz invariance is a continuous symmetry of a smooth manifold. The DAG does not have Lorentz invariance. The DAG has *causal invariance* (O02)[LV]: the quantum measure is independent of the linear extension of the partial order.

Three structural features of the DAG determine the emergent symmetry:

1. *Acyclicity* distinguishes one direction (the causal order) from the rest, giving signature $(3, 1)$ in $d = 4$.
2. *Causal invariance* ensures no preferred spacelike direction (discrete "no preferred frame").
3. $d = 4$ (L11) fixes the dimension.

The unique continuous symmetry group acting on a $(3, 1)$-signature 4-dimensional space with isotropy is $SO(3, 1)$—the Lorentz group. Lorentz invariance is what causal invariance looks like in the continuum approximation.

5.3 Gauge symmetries are emergent topology

The Standard Model gauge groups $SU(3)$, $SU(2)$, $U(1)$ are continuous Lie groups. In RA, they emerge from the discrete BDG topology types (Section 3.3). The gauge groups are *unitary* (not orthogonal) because the BDG amplitude $a(v|C) = e^{iS}$ is complex—a structural requirement for quantum interference. The maximum gauge group rank is 3 (the number of spatial dimensions in $d = 4$), which rules out Grand Unified Theory groups like $SU(5)$ or $SO(10)$.

The BDG closure theorem (L11) proves that no additional stable topology types exist in $d = 4$, which is why no beyond-Standard-Model gauge bosons have been observed and none are predicted.

6 Predictions

Because the constraint web has no free parameters, every node is either confirmed or falsified by experiment. The following predictions are specific, quantitative, and testable:

1. **BMV null result:** A mass in quantum superposition sources zero gravitational field. Strict null result for gravity-mediated entanglement experiments. *Near-term.*
2. **No WIMP signal:** No dark matter detection below the neutrino floor. WIMPs, axions, and sterile neutrinos are categorically excluded. *Ongoing (LZ, XENONnT).*
3. **Hubble gradient:** H_0 correlates linearly with baryon density along the line of sight. KBC void: $H_0 \approx 75.9\,\mathrm{km/s/Mpc}$. *Testable with DESI.*
4. **Spin-bath timescale:** $t^* \approx 0.274/g$ (parameter-free). Distinguished from GRW/CSL by the absence of free parameters. *Superconducting circuits.*
5. **Kinematic Coherence Bound:** $N_{\max} = \eta \cdot p_{\mathrm{th}}$ for fault-tolerant quantum arrays. Hard ceiling, not exponential suppression. *QEC at* 10^3 *logical qubits.*
6. **Neutrino mass sum:** $\Sigma m_\nu \approx 59\,\mathrm{meV}$. *CMB-S4 and Euclid.*
7. **Boltzmann Brain prohibition:** Structural, not statistical. *Cosmological consistency.*
8. $\Lambda = 0$ **exactly:** The cosmological constant is identically zero. *Confirmed by current bounds.*
9. **Baryon number exactly conserved:** No proton decay. *Ongoing (Super-Kamiokande, Hyper-Kamiokande).*
10. **Causal Firewall biosignatures:** Life wherever F_1 (Quantum Darwinism redundancy) and F_2 (non-relativistic condition) are satisfied, regardless of chemistry. *Long-term (SETI).*

These predictions are not independent hypotheses. They are co-determined by the constraint web. Any single failure breaks the web.

7 The Web as a New Kind of Theory

The five constraint loops of Section 4 have a consequence that deserves explicit statement: RA is not a conventional physical theory in the sense of a deduction tree with axioms, theorems, and predictions arranged in a hierarchy. It is a self-consistent constraint web in which every major claim mutually constrains every other.

This has implications for how the theory should be evaluated:

- **No independent tuning.** There are no free parameters that can be adjusted to fit data. The coupling constants, the particle spectrum, the dimensionality, the actualization threshold, and the gauge groups are all co-determined. Changing any one changes all the others.
- **Dissolution, not solution.** Many of the "hard problems" in Section 1.1 are not solved by RA in the sense of finding an answer within the existing mathematical framework. They are *dissolved*: shown to be artifacts of the continuum approximation that do not exist in the discrete theory.
- **A single failure is fatal.** The rigidity that makes the web powerful—the absence of independently tunable parts—is the same rigidity that makes it vulnerable. A positive BMV result, a WIMP detection, or a Hubble tension that does not correlate with

baryon density would break the web, not just one prediction.

- **The web is the theory.** The theory is not any individual result. It is the constraint structure that connects them all. Any linear presentation—including this paper—necessarily misrepresents the theory by imposing a sequence on what is fundamentally a simultaneous system of mutual constraints. The full theory is the web, accessible interactively at `relationalactualism.org`.

This is a genuinely new kind of physical theory, and it requires a genuinely new kind of evaluation: not "is axiom A true?" but "is the web self-consistent, and do its predictions match observation?"

The web is self-consistent. Its predictions, to date, match observation. The search continues.

8 Formal Verification Status

The discrete graph-theoretic core has been formally verified in Lean 4 (Table 1). These are machine-checked mathematical facts about finite graphs and integer combinatorics; whether the universe is described by these structures is a separate question, but the mathematical facts are not in doubt.

Table 1: Lean 4 verified results (zero `sorry` tags).

Result	**File**	**Status**
Local Ledger Condition (L01)	`RA_D1_Proofs.lean`	Proved
Graph Cut Theorem (L02)	`RA_D1_Proofs.lean`	Proved
$\alpha_{\mathrm{EM}}^{-1} = 137$ (L08)	`RA_D1_Proofs.lean`	Proved
Koide $K = 2/3$ (L09)	`RA_D1_Proofs.lean`	Proved
Confinement depths (L10)	`RA_D1_Proofs.lean`	Proved
BDG closure, 5 types (L11)	`RA_D1_Proofs.lean`	Proved
Frame independence (L04)	`RA_AQFT_Proofs_v10.lean`	Proved
Rindler stationarity (L05)	`RA_AQFT_Proofs_v10.lean`	Proved
Amplitude locality (O01)	`RA_AmpLocality.lean`	Proved
Causal invariance (O02)	`RA_AmpLocality.lean`	Proved

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